Symbols used in the README of Pre-gravity6plane

symbol	definition
$x_0..x_7$	coordinates: hidden, space (1-3), time (4), superluminal time (5-7); the Pre-Universe part writes them x0, ..., x7
η	$\mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$
τ_A, $\bar{\tau}_A$	real 8x8 matrices of Pre-gravity6plane.nb, $\bar{\tau}_A = \sigma\tau_A^T\sigma$, $\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$
$T^A = (T16^A)[A]$	$\begin{pmatrix} 0 & \bar{\tau}_A \\ \tau_A & 0 \end{pmatrix}$, real Dirac matrices
S^{AB}	$\frac{1}{4}[T^A, T^B]$
Spin(4,4), Pin(4,4)	the spin group of η, whose Lie algebra is spanned by the S^{AB}; the pin group, generated by Spin(4,4) and the reflections T^C
Γ, $P_{L,R}$, σ_{16}	$T^0\cdots T^7$; $\frac{1}{2}(I_{16} \mp \Gamma)$; $T^0T^1T^2T^3$
ψ, $\bar{\psi}$	dirac16complex (complex Grassmann 16-spinor) $\psi = (\psi_0, \ldots, \psi_{15})$, $\bar{\psi} = \psi^\dagger\sigma_{16}$; in the Pre-Universe part, a real 16-component spinor $\psi \in \mathbb{R}^{16}$
$\psi^\dagger$, ψ^*	conjugate transpose and complex conjugate of ψ
ψ_1, ψ_2	type-1 and type-2 halves of ψ: components 0..7 and 8..15; in the Pre-Universe part, $\psi_1 = \psi_1^a$ and $\psi_2 = \psi_2^{(a)}$ $(a = 0, \ldots, 7)$, the $\overline{SO(4,4;\mathbb{R})}$ counterparts of the left-handed and right-handed spin(8) spinors, with $\psi \in \mathbb{R}^{16} = \{\psi_1 \in \mathbb{R}^8 \oplus \psi_2 \in \mathbb{R}^8\}$
∂_A, ∂_μ	$\partial/\partial x_A$, $\partial/\partial x_\mu$
$\mathcal{L}$, m, λ	Lagrangian, mass, quartic coupling; $\mathcal{L}_m$ and $\mathcal{L}_{-m}$ are $\mathcal{L}$ with mass parameter m and $-m$
K, K_{ab}, δ	$K = -i\sigma_{16}T^4$ with entries K_{ab}; $\{\psi_a(x), \psi_b^\dagger(y)\} = K_{ab}\delta(x-y)$ at equal x_4, with δ the Dirac delta of the other seven coordinates
$\psi_\pm$, $S_\pm$, $\mathcal{L}_D$	$K\psi_\pm = \pm\psi_\pm$; $S_\pm = \psi_\pm^\dagger(-iT^4)\psi_\pm$; $\mathcal{L}_D[\chi] = \chi^\dagger(-iT^4)(T^A\partial_A - m)\chi$
M, H	mass and Hubble parameters of Pre-gravity6plane.nb; $m = -2M$
Ψ_{16}	Universe wave function of Pre-gravity6plane.nb, $\Psi_{16} = (f16[0][x_0, x_4], \ldots, f16[15][x_0, x_4])$; the Pre-Universe part writes it [CapitalPsi]16, the 16-component $O(4,4)$ spinor of the Dirac equation for the Pre-Universe
z, t, p, q, u	$z = 6Hx_0$, $t = Hx_4$, mode $e^{pz+qt}u$ with u a constant spinor
${F_A}^a$, ${F_a}^A$, V_1, V_2	triality frames and vector forms: $V_1 = \eta(\psi_1^T\sigma{F_A}^a)$, $V_2 = {F_a}^A\psi_2$
n_k, Z, M^6	$n_k = V_k[\mathrm{Re}\,\psi]$; $Z \in \mathbb{R}^8$; $M^6 = \{Z : n_1 \cdot Z = n_2 \cdot Z = 0\}$
E, $\binom{8}{2}$	energy of a universe for translations of x_4 (a pair of universes has energies $\pm E$); binomial coefficient, $\binom{8}{2} = 28$
Re, Im	real and imaginary parts; for x in Os, Re $x = \langle x, 1\rangle$ and Im $x = x -$ Re x
f, r_s, H_Λ	$1 - r_s/x_0$ (modern), $1 - r_s/x_0 - H_\Lambda^2 x_0^2$ (future)
a_4	function in the vielbein of Pre-gravity6plane.nb; $a_4(Hx_4) = Hx_4$ for the primordial field
${e^A}_\mu$, $g_{\mu\nu}$, g, ds^2	vielbein; metric $g_{\mu\nu} = \eta_{AB}{e^A}_\mu{e^B}_\nu$; g its determinant, which vanishes on an Einstein-Rosen bridge; line element $ds^2 = g_{\mu\nu}dx_\mu dx_\nu$ (summed over $\mu, \nu = 0, \ldots, 7$)
${E_A}^\mu$, δ^μ_ν, ${\Gamma^\nu}_{\mu\lambda}$, ${\omega_\mu}^A{}_B$	inverse vielbein, ${E_A}^\mu{e^A}_\nu = \delta^\mu_\nu$ (Kronecker symbol); Christoffel symbols of $g_{\mu\nu}$; spin connection ${\omega_\mu}^A{}_B = {e^A}_\nu(\partial_\mu{E_B}^\nu + {\Gamma^\nu}_{\mu\lambda}{E_B}^\lambda)$
γ^μ, D_μ	${E_A}^\mu T^A$; $\partial_\mu + \frac{1}{4}{\omega_\mu}^A{}_B T_A T^B$
ρ_s, h_{KS}, α, β, p, B	Kohn-Sham density, $\alpha p + BM\beta$, $-T^4T^0$, $-iT^4$, $-i\,d/d\rho$, lapse e_4
ΔSCF	excited states found as self-consistent Kohn-Sham solutions with a changed occupation of the levels
$\Omega(T)$	Kohn-Sham grand potential at temperature T
R_1	$x_1 \mapsto -x_1$
R, κ, T_H, ϵ	scalar curvature, surface gravity, $\kappa/2\pi$, Killing energy
Kretschmann scalar	$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, with $R_{\mu\nu\rho\sigma}$ the Riemann tensor
I_n	$n \times n$ identity matrix
g_{jj}, b_{jj}, a	deflating metric functions $g_{jj} = b_{jj}e^{-ax_4}$ $(j = 5, 6, 7)$, $a > 0$, $b_{jj} > 0$
w, y_j	coordinates of the deflating block: $w = \frac{2}{a}e^{ax_4/2}$, $y_j = \sqrt{b_{jj}}\,x_j$
$\mathbb{R}^4$	flat 4-space of x_0, x_1, x_2, x_3
ℓ, p_5, x_4^*	affine parameter; $p_5 = -g_{55}\,dx_5/d\ell$; turning point $\frac{1}{a}\ln(b_{55}/p_5^2)$
ω, $k_0..k_3$, q_5, q_6, q_7	frequency and wave numbers of a universe $e^{i(k_0x_0+\cdots+q_7x_7-\omega x_4)}u$ (the indexed q_j are wave numbers, not the exponent q); u a constant spinor
a_c	critical deflation rate $2H$
χ, M, c_A	mode spinor with $\chi' = d\chi/dx_4 = M\chi$; $M = T^4(i\sum_{A\neq 4} c_A T^A - m)$; real coefficients c_A
$P_\pm$	$\chi^\dagger\frac{1}{2}(I_{16} \pm K)\chi$, weights of the +mass and -mass sectors
$\mathbb{R}^{4,4}$	real eight-dimensional pseudo-Euclidean vector space with the $SO(4,4;\mathbb{R})$-invariant quadratic form $x_0^2 + x_1^2 + x_2^2 + x_3^2 - x_4^2 - x_5^2 - x_6^2 - x_7^2$
$SO(4,4;\mathbb{R})$, $SO(4,4)$	real linear maps of $\mathbb{R}^{4,4}$ with determinant 1 that preserve this quadratic form

symbol	definition
$\overline{SO(4,4;\mathbb{R})}$, $\overline{SO(4,4)}$	universal cover of $SO(4,4;\mathbb{R})$; the Pre-Universe part writes Spin(4,4) (= $\overline{SO(4,4)}$)
Spin(3,4)	the subgroup of Spin(4,4) that preserves both $\mathbb{R}$ and Im Os; under it Ψ_1 and Ψ_2 transform the same way, and the index (a) is written a
SO(8), Spin(8), spin(8), D_4	rotation group of $\mathbb{R}^8$ and its double cover; spin(8) spinors are the spinors of Spin(8); D_4 is their Dynkin diagram: a central node (adjoint representation) joined to the vector node and to the left-handed (type 1) and right-handed (type 2) spinor nodes
$O_{4,4}$, Os	split octonion algebra: real eight-dimensional algebra with unit 1, a non-degenerate indefinite inner product $\langle\ ,\ \rangle : Os \times Os \to \mathbb{R}$ (written $<\ ,\ >$ in the README), and a multiplication $Os \times Os \to Os$ with $\langle xy, xy\rangle = \langle x,x\rangle\langle y,y\rangle$ and $\langle 1,1\rangle = 1$; $\mathbb{R}$ is identified with the subalgebra generated by 1
$Im\ Os$	orthogonal complement of 1 in Os; $Os = \mathbb{R} \oplus Im\ Os$
x^* (x in Os)	conjugate, $x^* = 2\langle x,1\rangle - x$; x lies in Im Os if and only if $x^* = -x$
$C_{4,4}$, id	(pseudo-) Clifford algebra generated by the $e_A^{4,4}$; id is its multiplicative identity
$e_A^{4,4}$	the eight generators of $C_{4,4}$, $A = 0,\ldots,7$; they anticommute and satisfy $e_A^{4,4}e_B^{4,4} + e_B^{4,4}e_A^{4,4} = id\ g_{AB}$, and the Pre-Universe part states that each has square $\pm$ id
g_{AB}, $g_{A,B}$	diag$(1,1,1,1,-1,-1,-1,-1)$
e_A, x^A	Minkowski vector basis of Os, $A = 0,\ldots,7$, with $e_0 = 1$ and $\langle e_A, e_B\rangle = g_{A,B}$; $x = x^A e_A$ with real x^A
m_{AB}^C	multiplication constants, $e_A e_B = e_C m_{AB}^C$, constructed from the irreducible representation of the $C_{4,4}$ generators and a type 1 spinor u with $\langle u,u\rangle = 1$; they give the split octonion multiplication table (row A, column B: $e_A e_B$)
$e_a^{1/2}$, $e_{(a)}^{1/2}$	type 1 and type 2 spinor bases of Os, $a = 0,\ldots,7$
Ψ_1, Ψ_2	type 1 and type 2 spinors in Os: $\Psi_1 = \psi_1^a e_a^{1/2}$ and $\Psi_2 = \psi_2^{(a)} e_{(a)}^{1/2}$, with real ψ_1^a and $\psi_2^{(a)}$
$\mathcal{C}_{2n}$	written Subscript[[ScriptCapitalC], 2 n]: complex Clifford algebra with $2n$ generators; Lord's reduced Brauer-Weyl generators form an irreducible representation of $\mathcal{C}_{2n}$ built from one of $\mathcal{C}_{2(n-1)}$
tau8, tau16	the real 8x8 and 16x16 matrices of Pre-gravity6plane.nb, obtained from Dirac's 4x4 gamma matrices with Lord's reduced Brauer-Weyl generators
r, 0, ø, t, u, m, g	in the passage quoted from Einstein and Rosen (1935): the coordinates r, θ, ϕ, t (printed r, 0, ø, t), the coordinate u with $u^2 = r - 2m$, the mass parameter m, and g, the determinant of the metric
$M_{4,4}$	Minkowski spacetime of signature {4,4} (in the notation that the README asks the reader to ignore for now)
o, eA, ea, ea[Psi]1, ea[Psi]2, [Sigma], [Eta], X, EA	the notation that the README asks the reader to ignore for now, in Mathematica form: o, an element of $O_{4,4}$, expanded in the vector basis eA (components o^j) and, by Cartan's triality, in the type 1 and type 2 spinor bases ea[Psi]1 and ea[Psi]2; after the restriction to Spin(3,4) both spinor bases are written ea; <Subscript[ea, a], Subscript[ea, b]> = Subscript[[Sigma], ab] and <Subscript[eA, A], Subscript[eA, B]> = Subscript[[Eta], AB]; {X, EA} lies in $M_{4,4} \otimes O_{4,4}$